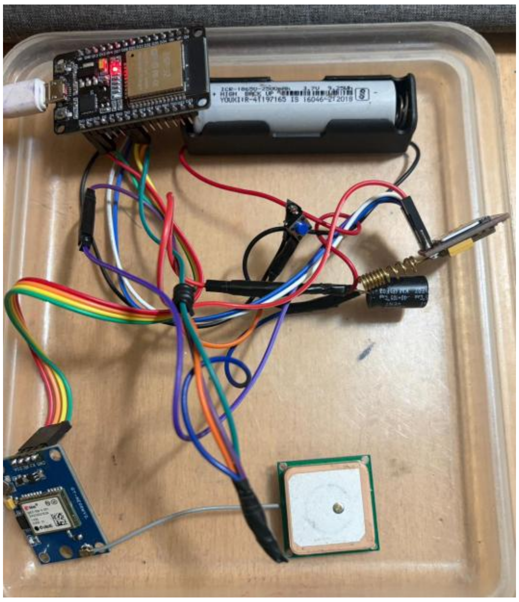
Hardware Setup

The hardware setup of the Kids Safety Tracker consists of multiple electronic components connected together to create a real-time child safety and emergency alert system.

The entire system is controlled by the ESP32 microcontroller, which acts as the brain of the project. It communicates with the GPS module, GSM module, SOS button, and Firebase cloud database to perform tracking and emergency alert operations.



Hardware Setup

1. ESP32 Dev Module

The ESP32 is the main controller of the system.

Functions of ESP32:
- Reads GPS coordinates from the NEO-6M GPS module
- Communicates with the SIM800L GSM module
- Detects SOS button presses
- Connects to WiFi
- Uploads location data to Firebase Realtime Database
- Controls SMS and calling operations

Why ESP32 is used:
- Built-in WiFi support
- Multiple UART serial ports

- Fast processing speed
- Low power consumption
- Suitable for IoT applications

In this project:
- UART2 is used for GPS communication
- UART1 is used for GSM communication

 # 2. NEO-6M GPS Module

The NEO-6M GPS module is responsible for obtaining the real-time location of the child.

Functions:
- Receives satellite signals
- Calculates latitude and longitude
- Sends GPS data to ESP32

Working:
- GPS continuously searches for satellites
- Once a valid GPS fix is obtained, coordinates are generated
- ESP32 reads these coordinates using TinyGPSPlus library
- A Google Maps link is created using latitude and longitude

Example:
https://www.google.com/maps?q=22.749500,75.846000

Connections:
- GPS TX → ESP32 GPIO16
- GPS RX → ESP32 GPIO17

 # 3. SIM800L GSM Module

The SIM800L GSM module provides GSM communication capabilities.

Functions:
- Sends emergency SMS alerts
- Makes automatic emergency calls
- Receives SMS commands like "LOC"

Working:
- ESP32 sends AT commands to SIM800L

- GSM module communicates through cellular network
- SMS with Google Maps location is delivered to parent phone

Emergency Process:
1. SOS button pressed
2. ESP32 fetches GPS location
3. SMS sent to parent
4. Automatic call initiated

Connections:
- GSM TXD → ESP32 GPIO26
- GSM RXD → ESP32 GPIO27

 # 4. SOS Push Button

The SOS button is used by the child during emergencies.

Functions:
- Triggers emergency alert sequence
- Sends location instantly to parent

Working:
- One side connected to GPIO14
- Other side connected to GND
- Uses INPUT_PULLUP configuration
- Normally reads HIGH
- Becomes LOW when pressed

When pressed:
- GPS location fetched
- SMS sent
- Call placed automatically

 # 5. 18650 Battery

The 18650 lithium-ion battery powers the portable system.

Functions:
- Supplies power to GSM module
- Makes device portable
- Allows operation without external power source

Importance:
- SIM800L requires stable high current
- Battery provides sufficient current during SMS/call transmission

Battery Specifications:
- 3.7V rechargeable lithium-ion battery
- Portable and lightweight

 # 6. Capacitor for Voltage Stabilization

A capacitor is connected across the GSM power supply.

Purpose:
- Stabilizes voltage during GSM transmission
- Prevents sudden voltage drops
- Avoids ESP32 restart problems

Why needed:
- SIM800L draws high current spikes during:
  - SMS sending
  - Calling
  - Network registration

Without capacitor:
- ESP32 may restart
- GSM module may fail

Typical capacitor:
- 1000µF electrolytic capacitor

 # 7. Working of Complete Hardware System

Step-by-step operation:

Step 1: Power ON
- ESP32 initializes all modules
- GSM connects to mobile network
- GPS starts searching satellites
- WiFi connects to Firebase

Step 2: GPS Tracking
- GPS module continuously sends coordinates
- ESP32 processes location data
- Firebase database updated

Step 3: Emergency Trigger
When child presses SOS button:
- ESP32 detects button press
- GPS location fetched
- Emergency SMS generated
- Google Maps link created
- SMS sent to parent
- Automatic call placed

Step 4: Remote Location Request
Parent sends:
LOC

Device replies with:
- Current GPS coordinates
- Google Maps link

Overall System Purpose

The hardware setup creates a low-cost, portable, real-time child safety device capable of:
- Emergency alerting
- Live location sharing
- Parent communication
- Cloud data storage

The system works even in low internet areas because GSM communication is used for emergency alerts.